# Detection of replay on Emmett's data with Linear Dynamical Systems

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## 1 Introduction

Here we use a Linear Dynamical System model (LDS)[1] to detect replay events in of Emmett's switching-task recordings (Thompson et al., 2024).

## 2 Methods

### 2.1 Data

We analyzed the spiking activity of all striatal neurons (neuron indices 0-207) in electrophysiological recording `EJT178_implant1/recording6_29032022`. Figure 1 plots a sample of spike times used in this analysis.

[Figure 1 about here.]

### 2.2 LDS model

The LDS model estimates $K$ latent variables, $x_0[n], \ldots, x_{K-1}[n]$, $n \in \{0, \ldots, N-1\}$. We concatenated the latent variables into a latent vector $\mathbf{x}[n]$:

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[1] https://github.com/joacorapela/ssm

$$\mathbf{x}[n] = \begin{bmatrix} x_0[n] \\ \vdots \\ x_{K-1}[n] \end{bmatrix} \tag{1}$$

The latents evolve in time following a first-order Markov process with additive Gaussian noise:

$$\mathbf{x}[n] = A\mathbf{x}[n-1] + \mathbf{v}[n] \quad \text{with } \mathbf{v}[n] \sim \mathcal{N}(0, Q)$$

starting from $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{x}_0|\mathbf{m}_0, V_0)$.

The latent variables are combined linearly to approximate the firing rate of the recorded neurons (bin size 200 ms), $\mathbf{y}[n]$, with a constant offset, $\mathbf{a}$, and an additive noise, $\mathbf{w}[n]$.

$$\mathbf{y}[n] = \mathbf{a} + C\mathbf{x}[n] + \mathbf{w}[n] \quad \text{with } \mathbf{w}[n] \sim \mathcal{N}(0, R)$$

We estimated an LDS models with $K = 5$ latent variables (Eq. 1; the selection of the number of latent variables was arbitrary, and we should perform model selection in later iterations).

We estimate the model parameter $\{\mathbf{m}_0, V_0, \mathbf{u}, A, Q, \mathbf{a}, Z, R\}$ using the expectation maximisation algorithm (doEstimateEM.py). We then used the Kalman Filter algorithm, with these parameters, to infer the latents variables, $\mathbf{x}[n]$ in Equation 1 (doKalmanFilteringWithEstimatedParams.py).

# 3 Results

[Figure 2 about here.]

[Figure 3 about here.]

[Figure 4 about here.]

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[Figure 6 about here.]

[Figure 7 about here.]

[Figure 8 about here.]

[Figure 9 about here.]

# References

Thompson, E., Rollik, L., Waked, B., Mills, G., Kaur, J., Geva, B., Carrasco-Davis, R., George, T., Domine, C., Dorrell, W., et al. (2024). Replay of procedural experience is independent of the hippocampus. *bioRxiv*, pages 2024–06.

# List of Figures


1 Example spikes times used in this analysis. We used recording session `EJT178_implant1/recording6_29-03-2022` and neurons indices 0 to 207 in the striatum. Click on the image to get its interactive version. . . . . 6

2 Orthonormalised latent variables ($\mathbf{x}(t)$ in Eq. 1) inferred from the data in the behavioral period used to estimate the models parameters (between 4045 and 4345 sec). The colored vertical lines indicate 'poke_in' times, with green, red, blue, yellow, purple, cyan and magenta corresponding to ports 1 to 7, respectively. The correct sequence is $2 \rightarrow 1 \rightarrow 6 \rightarrow 3 \rightarrow 7$ corresponding to red $\rightarrow$ green $\rightarrow$ blue $\rightarrow$ cyan $\rightarrow$ magenta. Click on the image to get its interactive version. . . . . 7

3 HMM emission parameters. Inferred means and standard deviations (error bars) of the Gaussian emission model for each latent (i.e., observation) and for each HMM state. . . . . 8

4 Inferred HMM transition matrix. . . . . 9

5 True (blue) versus estimated (red) behavioral states, when estimation and inference was done on the same data segment of the behavioral period. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from the same data segment. . . . . 10

6 Orthonormalised latent variables inferred from neural data in the behavioral period not used to train LDS or HMM model parameters (in the behavioral segment between 4343 and 4645 sec). 11

7 True (blue) versus estimated (red) behavioral states, when estimation and inference were done on different data segments of the behavioral period. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from data between 4345 to 4645 sec. . . . . 12

8 Orthonormalised latent variables inferred from data in a non-behavioral block immediately following behavior between 6000 and 7454 sec. . . . . 13

9 True (blue) versus estimated (red) behavioral states, when estimation was done with data in the behavioral period and inference was done with data in a non-behavioral period immediately following behavior. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from data between 6000 to 7454 sec. . . . . 14

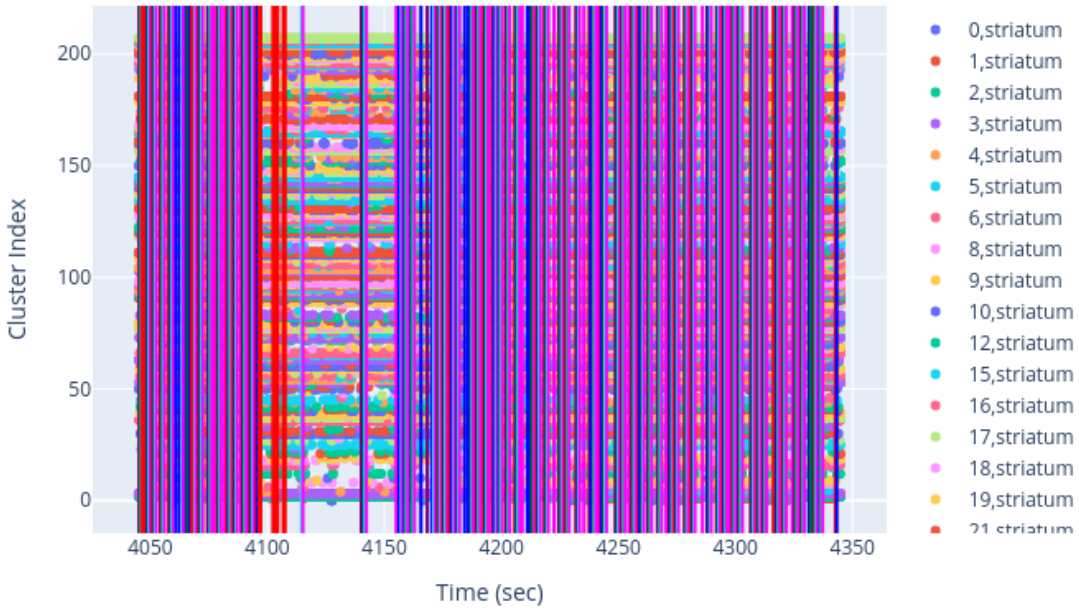



Figure 1: Example spikes times used in this analysis. We used recording session `EJT178_implant1/recording6_29-03-2022` and neurons indices 0 to 207 in the striatum. Click on the image to get its interactive version.

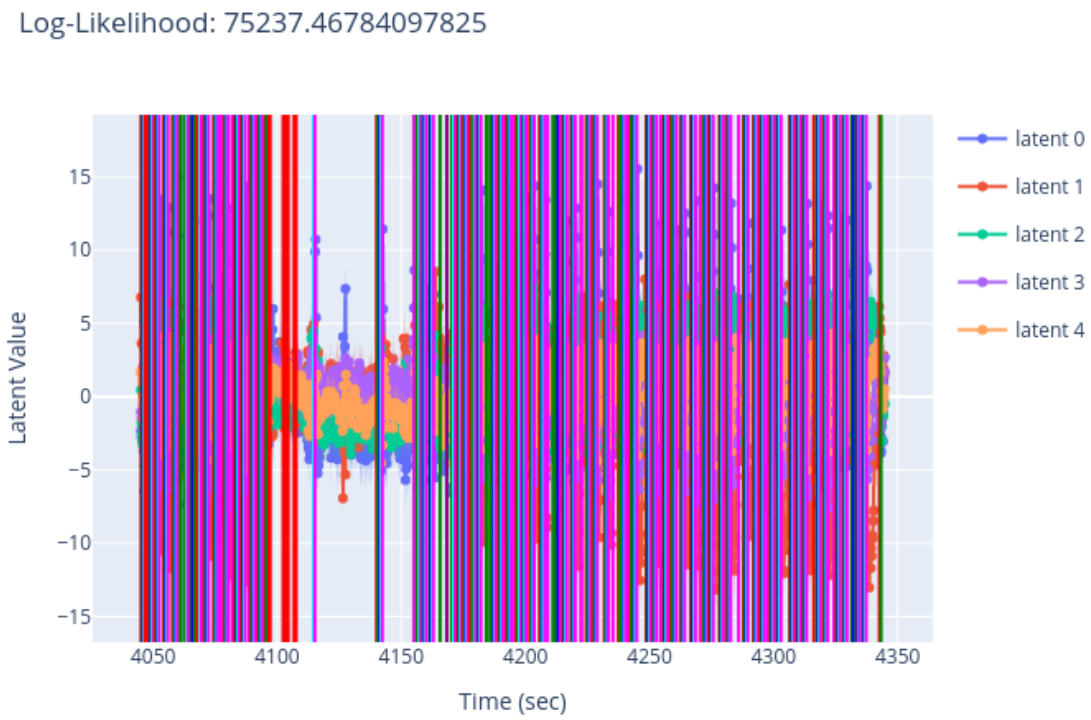



Figure 2: Orthonormalised latent variables ($\mathbf{x}(t)$ in Eq. 1) inferred from the data in the behavioral period used to estimate the models parameters (between 4045 and 4345 sec). The colored vertical lines indicate 'poke_in' times, with green, red, blue, yellow, purple, cyan and magenta corresponding to ports 1 to 7, respectively. The correct sequence is $2 \to 1 \to 6 \to 3 \to 7$ corresponding to red $\to$ green $\to$ blue $\to$ cyan $\to$ magenta. Click on the image to get its interactive version.

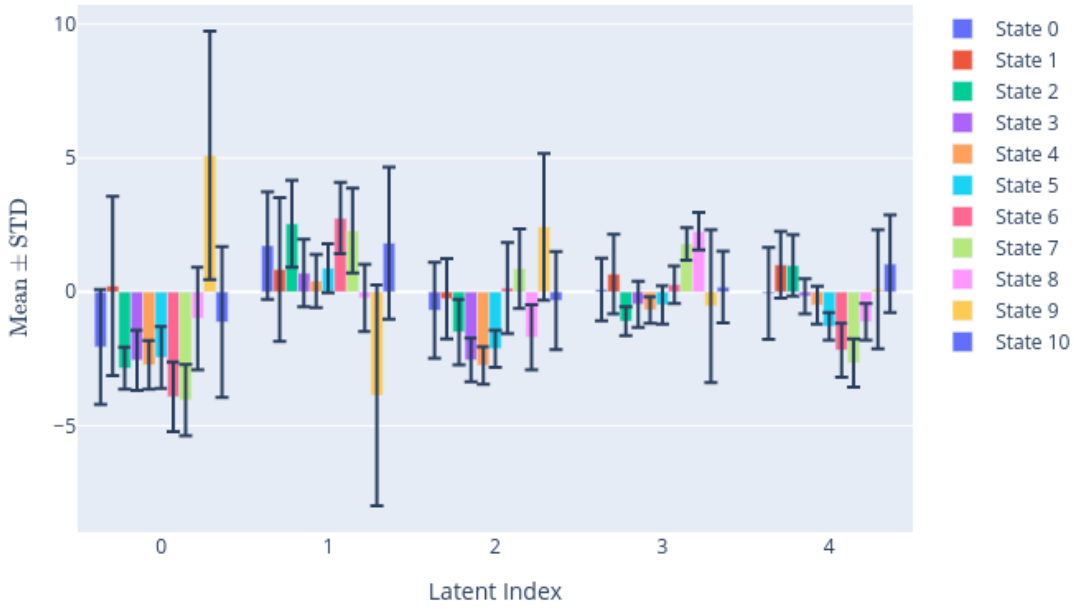



Figure 3: HMM emission parameters. Inferred means and standard deviations (error bars) of the Gaussian emission model for each latent (i.e., observation) and for each HMM state.

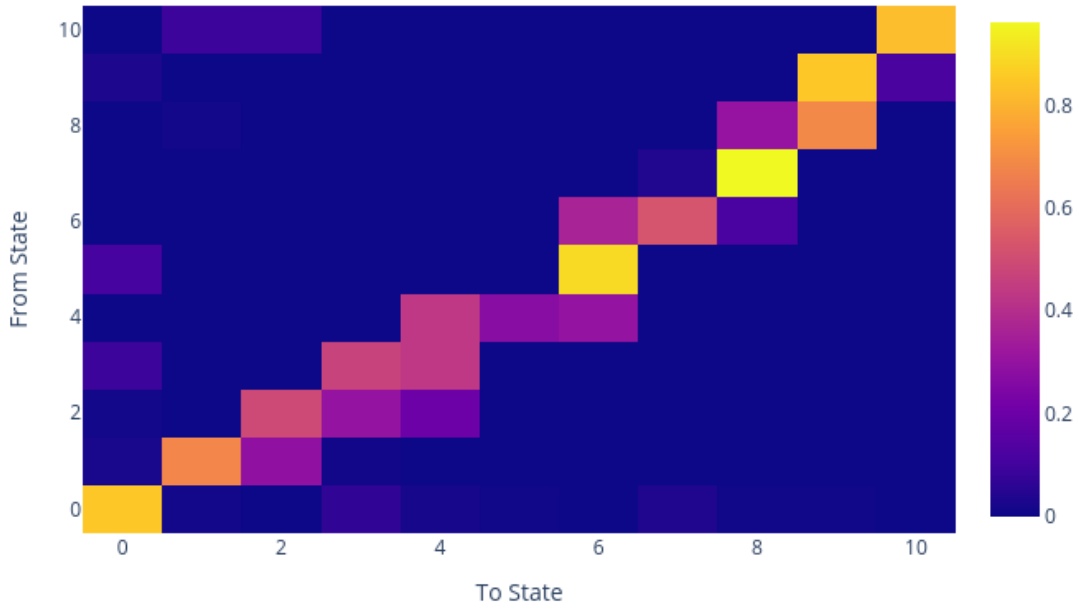



Figure 4: Inferred HMM transition matrix.

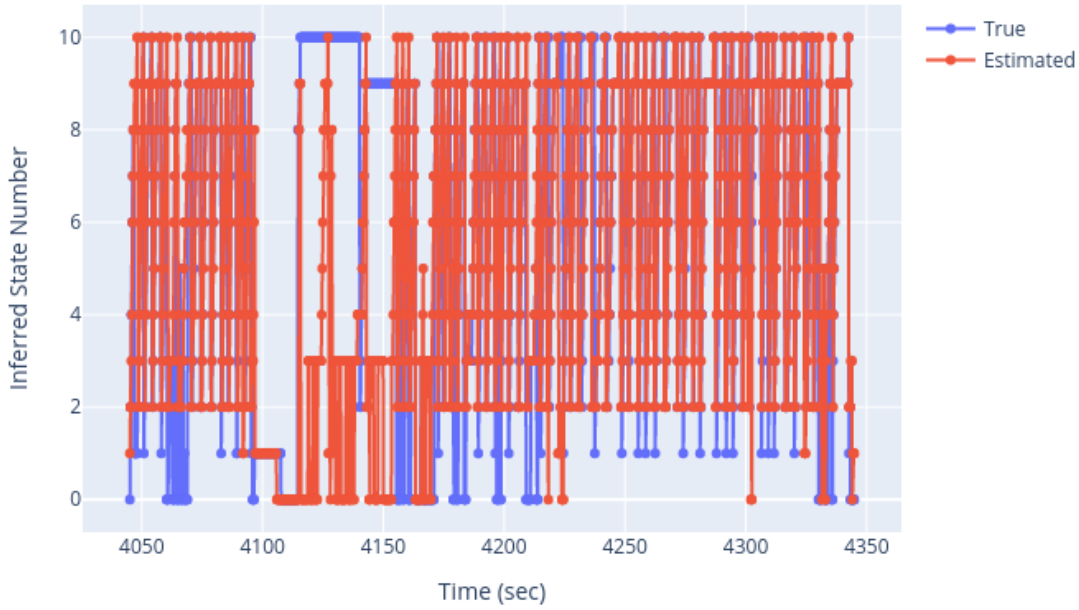



Figure 5: True (blue) versus estimated (red) behavioral states, when estimation and inference was done on the same data segment of the behavioral period. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from the same data segment.

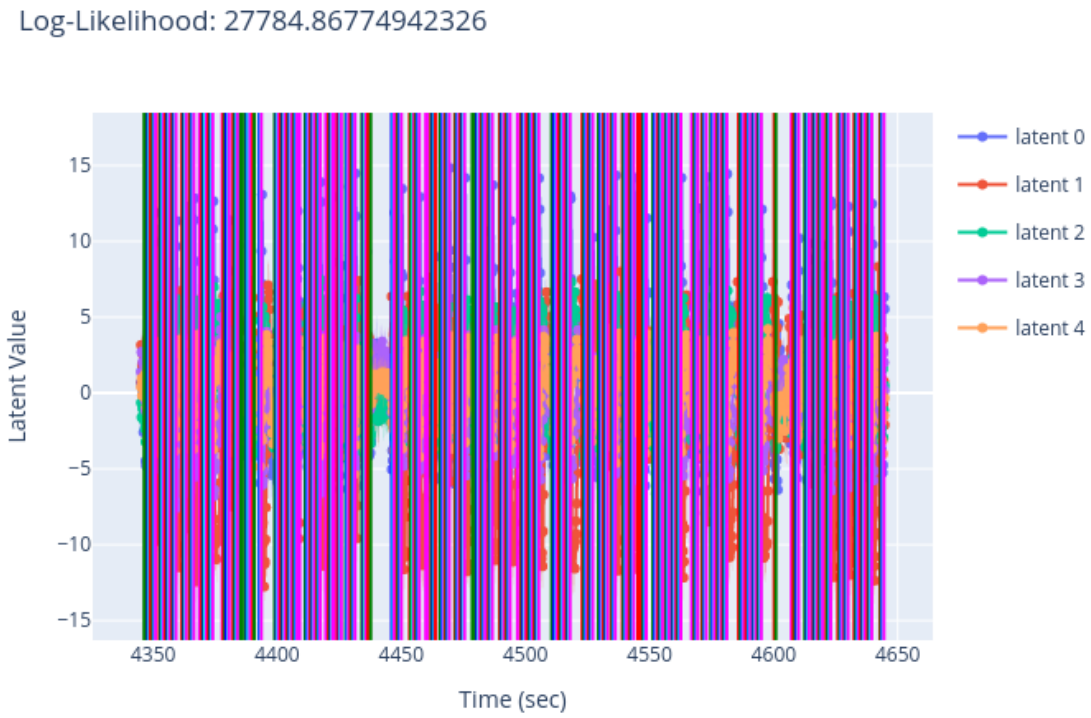



Figure 6: Orthonormalised latent variables inferred from neural data in the behavioral period not used to train LDS or HMM model parameters (in the behavioral segment between 4343 and 4645 sec).

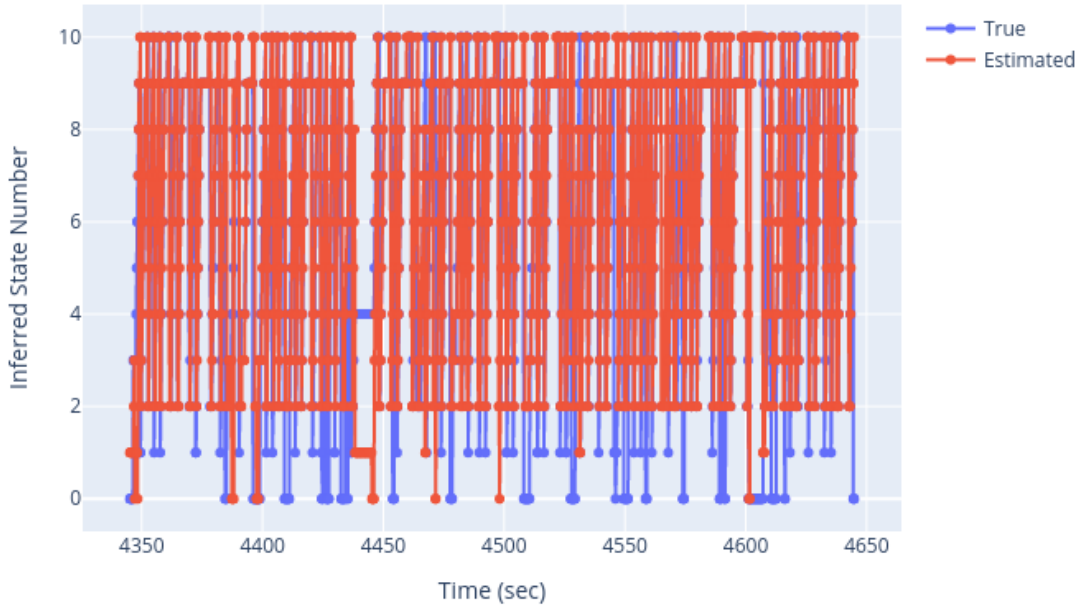



Figure 7: True (blue) versus estimated (red) behavioral states, when estimation and inference were done on different data segments of the behavioral period. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from data between 4345 to 4645 sec.

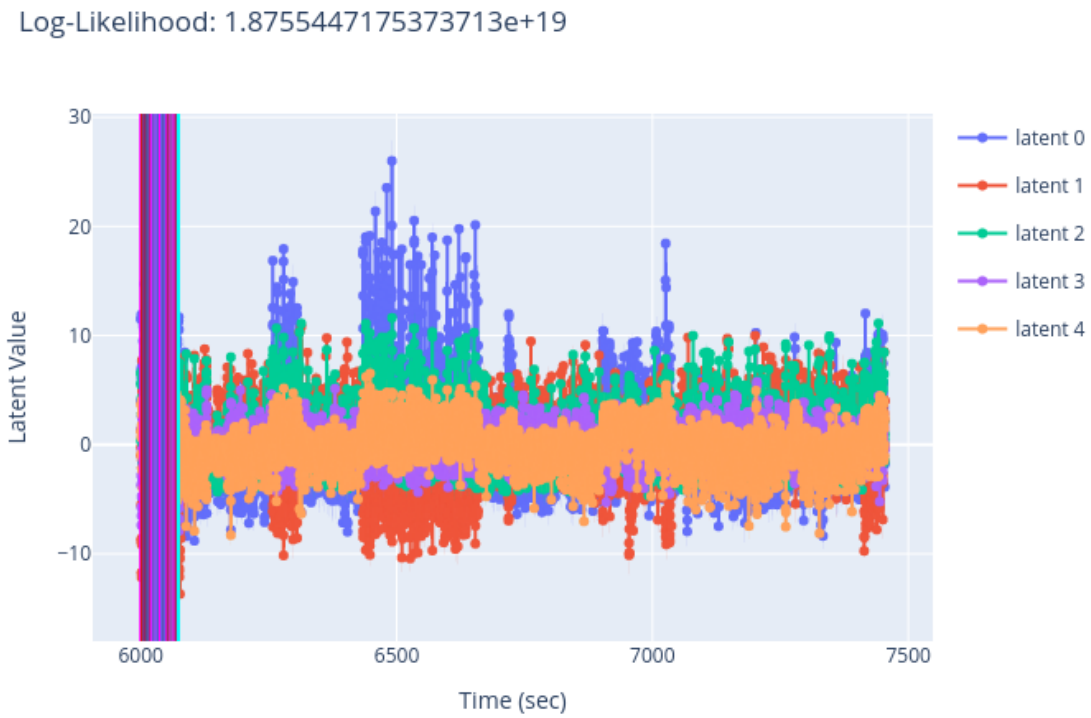



Figure 8: Orthonormalised latent variables inferred from data in a non-behavioral block immediately following behavior between 6000 and 7454 sec.

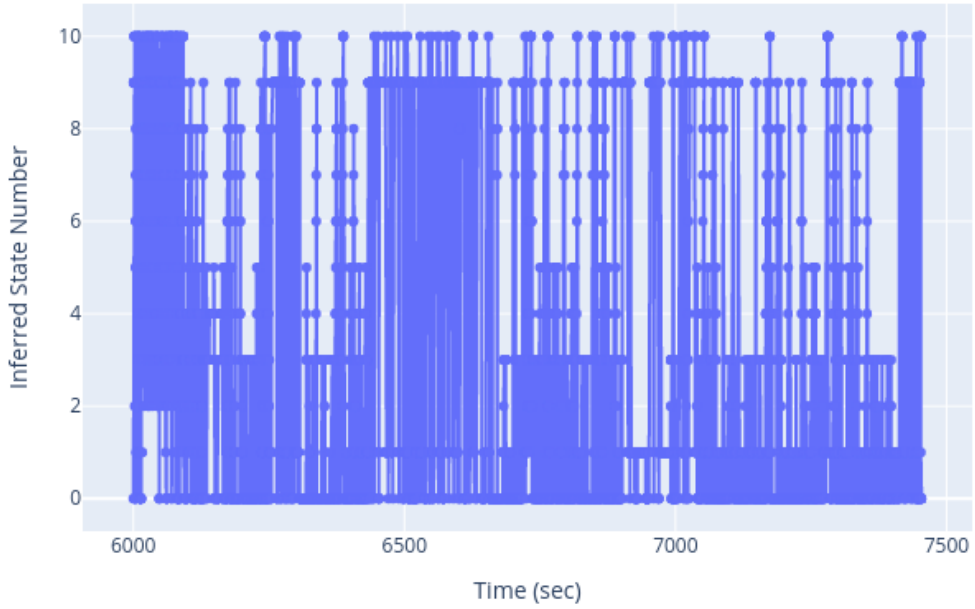



Figure 9: True (blue) versus estimated (red) behavioral states, when estimation was done with data in the behavioral period and inference was done with data in a non-behavioral period immediately following behavior. The LDS and HMM models were estimated with data between 4045 to 4345 sec. The LDS model inferred latents and the HMM inferred behavioral states from data between 6000 to 7454 sec.